

Assignment 4: Build a Semantic Search Ready NLP Pipeline from Raw Text Data

Business Scenario

A media company has thousands of customer reviews, blog comments, and article feedback messages. They want to:

- Clean noisy text data
- Standardize language using lemmatization
- Convert text into vector embeddings
- Find similar comments automatically
- Build an end-to-end NLP preprocessing system using NLTK, SpaCy, Scikit-learn and Gensim Model.

You must perform the following steps and ensure to create an end-to-end NLP model

1. Text cleaning
2. Tokenization
3. Stopword removal
4. Lemmatization
5. Vocabulary creation
6. Bag of Words / TF-IDF vectors
7. Word Embeddings
8. Sentence Embeddings
9. Similarity Search
10. Insights & Visualization

Dataset: Use HuggingFace dataset